

Katalin Berlow

Curriculum Vitae

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Research Interests

I have recently transitioned my career from Pure Mathematics research to more technical work. This summer, I graduated from UC Berkeley with my PhD studying Measured Group Theory. I left math with the aim of having a career with more tangible impact. My current research focuses on verifiable resource utilization for matrix multiplication.

Research & Employment

Member of Technical Staff — 0xPARC, San Francisco, CA

Aug 2026–Present

- Researching, in collaboration with with Daniel Reuter, **proof-of-resource-utilization protocols for matrix multiplication**, intended to let auditors verify execution of declared accelerator workloads even with adversarial inputs and implementations.
- We are in particular looking into protocols for proof of useful space.

Stephen E. Warschawski Visiting Assistant Professor

August 2026

University of California, San Diego, La Jolla, CA

- Postdoc researching measured group theory.
- Left position to pursue work in AI safety.

Research Intern, Cryptography — 0xPARC, San Francisco, CA

Jun–Aug 2026

- Designed and implemented CKKS-based fully homomorphic encryption algorithms for privacy-preserving fuzzy string matching, including insertion/deletion-tolerant scoring using ciphertext rotations and low-degree polynomial approximations.
- Designed a fully homomorphically encrypted k-nearest neighbors algorithm under CKKS.

Mathematical Problem Contributor — Epoch AI / FrontierMath

Spring 2025

- Contributed a graduate-level combinatorics problem to an AI reasoning benchmark, designed to be adversarially precise, resistant to shortcut solutions, and diagnostic of multi-step mathematical reasoning.

Education

Ph.D. in Mathematics

2021–2026

University of California, Berkeley, Berkeley, CA

Advisors: Andrew Marks and Robin Tucker-Drob

B.S. in Mathematics with Honors

2017–2021

Carnegie Mellon University, Pittsburgh, PA

Honors and Fellowships

Simons Dissertation Fellowship in Mathematics

2025

NSF Graduate Research Fellowship

2021

Fulbright Fellowship, *not completed*

2021

Benter Foundation Mathematics Scholarship

2021

Duluth REU

2020

Technical Skills & Additional

Programming: Python, NumPy, Git/GitHub, Linux/command line, SSH, L^AT_EX.

AI security and cryptography: proof of useful work, verifiable computation, AI compute governance, fully homomorphic encryption (CKKS).

Mathematics: combinatorics, graph theory, algorithms and complexity, probability, measure theory, dynamical systems, ergodic theory.

Research Publications

In Preparation

- [1] **A Measurable Analog to Euler's Formula in pmp Graphs** (in preparation).
- [2] **Equivalence Relations Acting on Bundles of Trees** (in preparation).
with R. Chen, A. Tserunyan, and R. Tucker-Drob
- [3] **Separating Complexity Classes of LCL Problems on Grids, Part 2** (in preparation).
with A. Bernshteyn, C. Lyons, and F. Weilacher

Submitted

- [4] **The Polychromatic Number of Grids** (submitted, 2025).
<https://arxiv.org/abs/2508.18559>. *with E. Hou*
- [5] **Separating Complexity Classes of LCL Problems on Grids** (submitted, 2025).
<https://arxiv.org/abs/2501.17445>. *with A. Bernshteyn, C. Lyons, and F. Weilacher*

Published

- [6] **Intersection Bodies of Polytopes**, *Beiträge zur Algebra und Geometrie* 63 (2022), 419–439.
<https://arxiv.org/abs/2110.05996>. *with M. Brandenburg, C. Meroni, and I. Shankar*
- [7] **Analysis on Almost Abelian Lie Groups**, *Journal of Mathematical Sciences* (2022).
<https://arxiv.org/abs/2004.04369>. *with M. Almora-Rios et al.*
- [8] **Restricted Stacks as Functions**, *Discrete Mathematics* 334 (2021).
<https://arxiv.org/abs/2008.01164>

Selected Talks

A Measurable Analog to Euler's Formula in pmp Graphs Joint Mathematics Meetings (scheduled)	Jan 2027
Workshop on Group Actions and Equivalence Relations Rényi Institute, Budapest, Hungary	Jul 2026
Local Problems on Grids: FIID/FFIID Separations and a Failure of Baire Shift Universality Caltech Logic Seminar	Apr 2026
Measurable Analog of Euler's Formula for pmp Graphs UT Austin Groups and Dynamics Seminar, Austin, TX	Mar 2026
A Measurable Analog to Euler's Formula Rényi Borel Combinatorics Seminar, Budapest, Hungary	Feb 2026
Separating Complexity Classes of LCLs on Grids North American Descriptive Set Theory Conference, University of Maryland	Oct 2025
Separating Complexity Classes of LCLs on Grids CanaDAM Combinatorics Conference, Ottawa, Canada	May 2025
Separating Complexity Classes of LCLs on Grids South Eastern Logic Symposium, University of Florida	Mar 2025